RBAC/security domains/local users	Not present in local current implementation	Implemented in remote Phase F
COOP/ISIS/Pod Policy Groups	Not present locally	Implemented in remote Phase E
Fault lifecycle/syslog monitoring	Not present locally	Implemented in remote Phase G
EVPN domain	Implemented locally and live-verified	Implemented and live-verified

LOCAL TERRAFORM REFERENCES

Primary module:

- [platform/terraform/aci/main.tf](../../platform/terraform/aci/main.tf)
- [platform/terraform/aci/providers.tf](../../platform/terraform/aci/providers.tf)
- [platform/terraform/aci/variables.tf](../../platform/terraform/aci/variables.tf)

The local ACI module currently contains 52 Terraform resource blocks. Relevant resource families include:

- Core: aci_tenant, aci_vrf, aci_bridge_domain, aci_subnet, aci_application_profile, aci_application_epg
- Policy: aci_filter, aci_filter_entry, aci_contract, aci_contract_subject, aci_epg_to_contract
- Logical L3Out: aci_l3_outside, aci_external_network_instance_profile, aci_l3_ext_subnet
- Access: aci_vlan_pool, aci_ranges, aci_physical_domain, aci_attachable_access_entity_profile, aci_leaf_access_port_policy_group
- IPG policies: aci_cdp_interface_policy, aci_lldp_interface_policy, aci_link_level_interface_policy, aci_spanning_tree_interface_policy, aci_port_security_interface_policy
- Physical access: aci_leaf_interface_profile, aci_access_port_selector, aci_epg_to_domain, aci_epg_to_static_path
- L3Out protocol path: aci_logical_node_profile, aci_logical_node_to_fabric_node, aci_logical_interface_profile, aci_l3out_path_attachment, aci_l3out_floating_svi, aci_l3out_bgp_protocol_profile, aci_bgp_peer_connectivity_profile, aci_l3out_ospf_interface_profile, aci_ospf_interface_policy
- VMM/L4-L7: VMM domain/controller/credential resources, L4-L7 device/interface, service graph, logical contexts, redirect policies/destinations
- VRF leaks: aci_vrf_leak_epg_bd_subnet

The local provider is pinned to CiscoDevNet/aci 2.20.0 in providers.tf and all new resource fields were checked against its local schema.

LOCAL MCP REFERENCES

Primary files:

- [mcp-server/src/mcp_server/tools/aci.py](../../mcp-server/src/mcp_server/tools/aci.py)
- [mcp-server/src/mcp_server/schemas/aci.py](../../mcp-server/src/mcp_server/schemas/aci.py)
- [mcp-server/src/mcp_server/clients/nautobot.py](../../mcp-server/src/mcp_server/clients/nautobot.py)

Local feature families include:

- Core: create_tenant, create_vrf, create_bridge_domain, create_epg
- Policy/L3Out: create_contract, create_l3out, create_vrf_route_leak

- Access: create_vlan_pool, create_physical_domain, create_aaep, create_leaf_interface_policy_group, create_leaf_interface_profile, create_interface_selector, create_static_path_binding, bind_epg_to_physical_domain, bind_epg_to_vmm_domain
- L3Out protocols: create_bgp_l3out, create_ospf_l3out, create_l3out_node_profile, create_l3out_interface_profile, create_l3out_interface, create_bgp_peer, create_ospf_interface, create_ospf_interface_policy
- VMM: create_vmm_domain
- L4-L7: create_l4l7_device, create_service_graph, create_one_arm_service_graph, create_pbr_policy, create_pbr_contract
- Status: show_status

Remote MCP comparison: the inspected remote branch explicitly includes refined create_vlan_pool, create_physical_domain, create_aep, create_leaf_interface_policy_group, bind_epg_domain, create_security_domain, and create_local_user APIs. Its access schemas use append/update semantics and domain-type-aware EPG bindings. The local names are compatible in intent but should not be blindly copied because the local worktree has additional L3Out/L4-L7 behavior and a different schema evolution.

UNIT-TEST REFERENCES

Local test files:

- [tests/unit/test_transformer.py](../../tests/unit/test_transformer.py): generator regression and emitted YAML coverage, including VRF/BD depth, EPGs, contracts, L3Outs, access policies, VMM, L4-L7, and route leaks.
- [mcp-server/tests/unit/test_schemas_aci.py](../../mcp-server/tests/unit/test_schemas_aci.py): Pydantic validation for ACI MCP requests.
- [mcp-server/tests/unit/test_tools_aci.py](../../mcp-server/tests/unit/test_tools_aci.py): tool-to-client argument/response behavior.
- [mcp-server/tests/unit/test_registry.py](../../mcp-server/tests/unit/test_registry.py): catalogue and schema-signature guarantees.
- [tests/unit/test_evpn_transformer.py](../../tests/unit/test_evpn_transformer.py) and EVPN MCP tests: second-domain regression coverage.

The last local focused validation recorded during this work was 25 transformer tests and 39 MCP schema/tool/registry tests passing. The local combined access/L3Out plan validated 22 to add, 0 to change, 0 to destroy; the OSPF policy plan validated 23 to add, 0 to change, 0 to destroy; the IPG policy plan validated 27 to add, 0 to change, 0 to destroy.

Remote test references observed in the repository tree and raw files:

- tests/unit/test_transformer.py: broad transformer coverage, including access policies, VMM, COOP/ISIS, Pod Policy Groups, fault/syslog policies, and AAA policy emission.
- mcp-server/tests/unit/test_tools_aci.py: tool-level tests for VLAN pools, Physical Domains, AAEPs, IPGs, VMM, domain binding, Security Domains, and Local Users.
- mcp-server/tests/unit/test_schemas_aci.py: request validation for the expanded remote tool catalogue.
- mcp-server/tests/unit/test_registry.py: MCP registry/signature checks.
- EVPN transformer, MCP, integration, and pyATS-equivalent tests remain present in both project lines.

RECOMMENDED RECONCILIATION

1. Treat the remote branch as the reference for Phase E/F/G ACI coverage: COOP/ISIS, RBAC/security domains/local users, and fault/syslog monitoring.
2. Preserve the local L3Out protocol, VRF-leak, L4-L7/PBR, one-arm service graph, OSPF-policy, and lab-template work unless the remote branch has a newer equivalent after direct file comparison.
3. Compare schema/API changes before any merge. In particular, remote bind_epg_domain is a consolidated physical/VMM operation, while the local checkout currently has separate bind_epg_to_physical_domain and bind_epg_to_vmm_domain tools.
4. Reconcile generated YAML schema changes before combining Terraform modules; do not merge main.tf mechanically.
5. Run both repositories' unit suites and Terraform validation after reconciliation.
6. Perform APIC-backed plans with temporary state before any apply.

LIMITATIONS

This report is based on read-only inspection of the public remote repository page/raw files and the current local worktree. It does not claim a commit-level semantic diff of every file. No remote code was fetched, merged, or copied. The remote README contains older operational values in places; use the local repository's current administration guide and isolated-stack configuration for local credentials and endpoints.